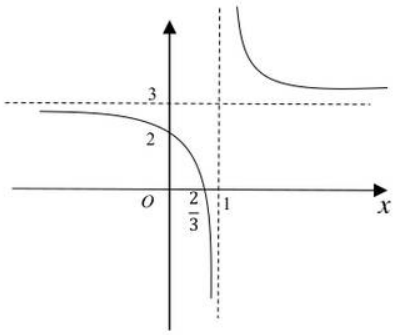
Type MS · Where a line meets a fraction curve again

Answer 1 of 4 · Question 9 part c · 4 marks

2020 JAN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

9(a)(i)	$\frac{2}{p} = 2$ so $p = 1$ *	M1A1cso
(ii)	$\frac{dy}{dx} = \frac{q(x-1)-(qx-2)}{(x-1)^2}$	M1A1
	When $x = 0$ $\frac{dy}{dx} = \frac{-q+2}{1} = -1, \Rightarrow q = 3$	M1A1,A1 (7)
(b)		B1ft B1ft B1 B1ft B1ft (5)
(c)	$x + 2 = \frac{3x-2}{x-1}$	M1
	$x^2 - 2x = 0$	M1
	$x(x-2) = 0$ $x = 2$	dM1A1cao(4) **[16]**

(a)(i)M1	Set $x = 0$ in the curve equation and equate result to 2. Obtain a value for p.
A1cso	Correct value of p obtained from a correct equation.
(ii)M1	Attempt the quotient rule. (formula is given on formula page). Denominator must be $(x-1)^2$. Numerator to be $q(x-1)-(qx-2)$ or $(qx-2)-q(x-1)$ **Must use $p = 1$ now or later.**
A1	Fully correct derivative
ALT	Use product rule: $\frac{dy}{dx} = q(x-1)^{-1} - (qx-2)(x-1)^{-2}$ M1 for attempt with 2 terms similar to above, either term to be correct A1 Both terms correct
M1	Set $x = 0$ in their derivative and equate to -1
A1	Correct equation
A1	$q = 3$

Type MS · Where a line meets a fraction curve again

Answer 1 of 4 · Question 9 part c · 4 marks

2020 JAN · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Question Number	Scheme	Marks
(b)	**No value for *q***: B0B1B0B1B1 available. **Incorrect *q*:** B1B1B0B1B1 available.	
B1ft	Equations of asymptotes seen or lines parallel to axes passing through $x = 1$, $y = 3$ drawn. $y = 3$ or their q. **Must have a value for *q*.**	
B1ft	Coordinates of crossing points seen explicitly or marked on the sketch. Must have $y = 2$; may have $x = 2 / q$ (value for q not needed)	
B1	Two branches in the correct "quadrants" Must have $q = 3$ for this mark. Asymptotes drawn.	
B1ft	There must be at least one branch of the curve drawn and 2 asymptotes drawn and labelled on the diagram by showing the coords of the points where they cross the axes or with their equations. The curve must not touch (or cross) either asymptote. ft their asymptotes**, inc *y* = *q***	
B1ft	Both crossing points clearly marked on their diagram. ft their crossing points .	
(c)		
M1	Eliminate y between the line and the curve equation. May use q or their value for q	
M1	Obtain a 2 or 3 term quadratic. May use q or their value for q.	
dM1	Solve their equation to obtain 1 or 2 values of x Depends on both M marks above.	
A1cao	$x = 2$ from a correct equation. If $x = 0$ is seen it must be clear that $x = 2$ is the only answer If x is eliminated: M1 elimination M1 obtain quadratic in y M1 solve for y A1 complete to a single value of x	

Type MS · Where a line meets a fraction curve again

Answer 1 of 4 · Question 9 part c · 4 marks

2020 JAN · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

Question Number	Scheme	Marks
	$\frac{dV}{}$	

Type MS · Where a line meets a fraction curve again

Answer 2 of 4 · Question 7 part c · 6 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

7(a)	$x = -\frac{5}{4}$	B1 [1]
(b)	$\{\frac{dy}{dx}\} = \frac{2x(4x+5) - 4(x^2-1)}{(4x+5)^2}$ $y = 0$ $\{\frac{dy}{dx}\} = \frac{2(-1)(4(-1)+5) - 4((-1)^2-1)}{(4(-1)+5)^2} = -2$ $\rightarrow \{gradient\ of\ normal =\} \frac{1}{2}$ $y - 0 = \frac{1}{2}(x-(-1))oe$	M1A1A1 B1 M1 M1A1cso [7]
(c)	$\frac{x+1}{2} = \frac{x^2-1}{4x+5}$ $2x^2 + 9x + 7 = 0$ ***OR*** $(x+1)(4x+5) = 2(x+1)(x-1)$ $(2x+7)(x+1) = 0 \Rightarrow x = \cdots$ ***OR*** $(4x+5) = 2(x+1) \Rightarrow x = \cdots$ $x = -\frac{7}{2}, \{x = -1\}$ $y = \frac{"-\frac{7}{2}"+1}{2} = \cdots$ $\left(-\frac{7}{2}, -\frac{5}{4}\right)$	M1 dM1 ddM1 A1 M1 A1 [6]
		Total 14 marks

Type MS · Where a line meets a fraction curve again

Answer 2 of 4 · Question 7 part c · 6 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	B1	$x = -\frac{5}{4}$ *or* $x = -1.25$, do not accept 4*x*+5=0
(b)	M1	Attempts to differentiate *y*. **Quotient Rule:** Look for $\frac{Px(4x+5) - Q(x^2-1)}{(4x+5)^2}$, *where* $P > 0, Q > 0$ If the quotient rule is quoted, it must be correct. Condone invisible bracket for this mark. **Product Rule:** Look for $Px(4x+5)^{-1} \pm Q(x^2-1)(4x+5)^{-2}$, *where* $P > 0, Q > 0$
	A1	Quotient rule: Correct denominator $(4x+5)^2$and one correct term in the numerator, 2*x*(4*x*+5) or $-4(x^2-1)$. Product rule: One term correct
	A1	Fully correct. $\left\{\frac{dy}{dx} =\right\} \frac{2x(4x+5)-4(x^2-1)}{(4x+5)^2}$ or $\left\{\frac{dy}{dx} =\right\} 2x(4x+5)^{-1} - 4(x^2-1)(4x+5)^{-2}$oe
	B1	*y* = 0 seen or implied
	M1	Correct method finding gradient of normal to the curve where *x*= -1
	M1	Correct method of forming the equation of a straight line using the point (**-1,their y value**) with a "**changed**" gradient.("changed" means a different gradient from their tangent gradient, their normal could be found by an incorrect method) If using *y*=*mx*+*c*, a value for *c* must be found and the equation formed.
	A1cso	For the correct line in any form, simplified or unsimplified. (isw) Once correct equation is seen, award the mark, ignore later incorrect simplification. e.g.2*y* = *x*+1 or 2*y*-*x*-1=0 or *y*=0.5x+0.5 oe or even $y - 0 = \frac{1}{2}(x - (-1))$
(c)	M1	For equating the equation of their line with the equation of the curve.
	dM1	For forming an equation with all like terms collected. **OR** cross multiplies then factorises e.g. $2(x^2 - 1) = 2(x+1)(x-1)$
	ddM1	Solves their equation with a correct method and finds at least one real value of *x* which is not -1, method of solving could be implied by correct answers if calculator used. **OR** cancels (*x*+1) and solves their linear equation.
	A1	$x = -\frac{7}{2}$, $\{x = -1\}$
	M1	For substituting their *x* value (not -1) to either the equation of their normal or the equation of the given curve to find a value for *y*
	A1	For the correct exact coordinates of $D\left(-\frac{7}{2}, -\frac{5}{4}\right)$ *oe* Allow *x* = -3.5, y = -1.25 instead of given as coordinates.

Type MS · Where a line meets a fraction curve again

Answer 2 of 4 · Question 7 part c · 6 marks

2024 JUN · Paper 02 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Where a line meets a fraction curve again

Answer 3 of 4 · Question 11 part d · 4 marks

2022 JUN · Paper 02 · extract 1 of 4 · find part d in this mark scheme extract

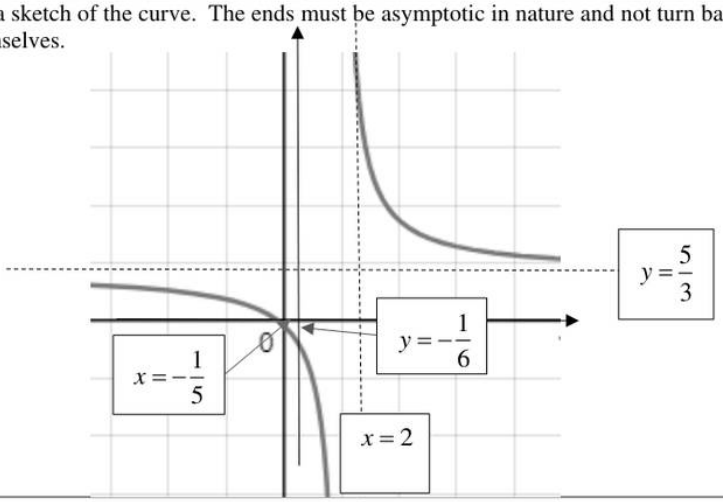
11 a	$\frac{dy}{dx} = \frac{(2a-1)(ax-6) - a((2a-1)x+1)}{(ax-6)^2}$	M1A1A1 (3)
b (i)	At $x = 0$ $\frac{dy}{dx} = -\frac{11}{12}$	B1
	So $\frac{dy}{dx} = \frac{-6(2a-1)-a}{36} = -\frac{11}{12}$	M1
	$-13a+6 = -33 \Rightarrow a = 3$*	dM1 A1 cso (4)
b (ii)	$y = \frac{5x+1}{3x-6}$*	B1 cso (1)
c	Asymptote with equation drawn at $y = 2$	B1
	Asymptote with equation drawn at $x = \frac{5}{3}$	B1
	Correct curve drawn with two branches	B1
	$-\frac{1}{5}$ and $-\frac{1}{6}$ labelled on x and y axes	B1 B1 (5)
d	$\frac{12}{11}x - \frac{1}{6} = \frac{5x+1}{3x-6}$	M1
	$(3x-6)(72x-11) = 66(5x+1)$	M1
	$x(72x-265) = 0$ oe	M1
	$x = \frac{265}{72}$ oe	A1 (4)
		Total 17 marks

Part	Mark	Notes
(a)	**M1**	For attempting to use a correct quotient rule: The definition of an attempt is as follows: • The denominator must be correct • The terms in the numerator must be subtracted either way around • The attempt to differentiate both terms must be as follows: $(2a-1)x+1 \to \pm(2a-1)$ and $ax-6 \to \pm a$
	A1	For one term in the numerator correct
	A1	For a fully correct differentiated expression. $\frac{dy}{dx} = \frac{(2a-1)(ax-6) - a((2a-1)x+1)}{(ax-6)^2}$ No simplification is required.

Type MS · Where a line meets a fraction curve again

Answer 3 of 4 · Question 11 part d · 4 marks

2022 JUN · Paper 02 · extract 2 of 4 · find part d in this mark scheme extract

	Uses Product Rule	
	M1	For attempting to use a correct product rule: The definition of an attempt is as follows: • The terms in the numerator must be added. • The attempt to differentiate both terms must be as follows: $(2a-1)x+1 \rightarrow \pm(2a-1)$ and $(ax-6)^{-1} \rightarrow \pm a(ax-6)^{-2}$
	A1	For one term correct
	A1	For a fully correct differentiation $\frac{dy}{dx}=\left[(2a-1)x+1\right](a)(ax-6)^{-2}+(ax-6)^{-1}(2a-1)$
(b)(i)	**B1**	For finding the gradient of the tangent $=-\frac{11}{12}$
	M1	For using that at the point A $x=0$ and setting their $\frac{dy}{dx}=\left(-\frac{11}{12}\right)$ which must be the gradient of their tangent to obtain $\frac{-6(2a-1)-a}{36}=\text{'}-\frac{11}{12}\text{'} \Rightarrow \left(\frac{-13a+6}{36}=-\frac{11}{12}\right)$
	dM1	For simplifying to a linear equation and attempting to solve.
	A1	For $a=3$ with no errors seen
(b)(ii)	**B1**	For $y=\frac{5x+1}{3x-6}$ exactly in that form. You must see evidence of substitution of $a=3$ for this mark. There must be $y=$ …. [Condone missing $x \neq 2$]
(c)		This is a sketch of the curve. The ends must be asymptotic in nature and not turn back on themselves. 
	B1	For asymptote **with equation drawn** at $y=\frac{5}{3}$ OR $x=2$
	B1	For both asymptotes **with equations drawn** at $y=\frac{5}{3}$ AND $x=2$
	B1	Correct curve drawn with two branches with one branch in 1st Quadrant and another branch in Quadrants 2, 3 and 4 with the graph drawn the correct way around. The graph must be asymptotic in nature.

Type MS · Where a line meets a fraction curve again

Answer 3 of 4 · Question 11 part d · 4 marks

2022 JUN · Paper 02 · extract 3 of 4 · find part d in this mark scheme extract

	B1	$-\frac{1}{5}$ labelled on x-axis **or** $-\frac{1}{6}$ labelled on y- axis
	B1	$-\frac{1}{5}$ labelled on x axis **and** $-\frac{1}{6}$ labelled on y- axis
(d)	**M1**	For setting $\frac{12}{11}x-\frac{1}{6}=\frac{5x+1}{3x-6}$ [eliminating y from both equations]
	M1	For forming a quadratic equation, 2TQ or 3TQ in any order. $(3x-6)(72x-11)=66(5x+1) \Rightarrow 72x^2-265x=0$
	M1	For solving their QE provided it is either a 2TQ or a 3TQ
	A1	For $x=\frac{265}{72}$ oe

Type MS · Where a line meets a fraction curve again

Answer 3 of 4 · Question 11 part d · 4 marks

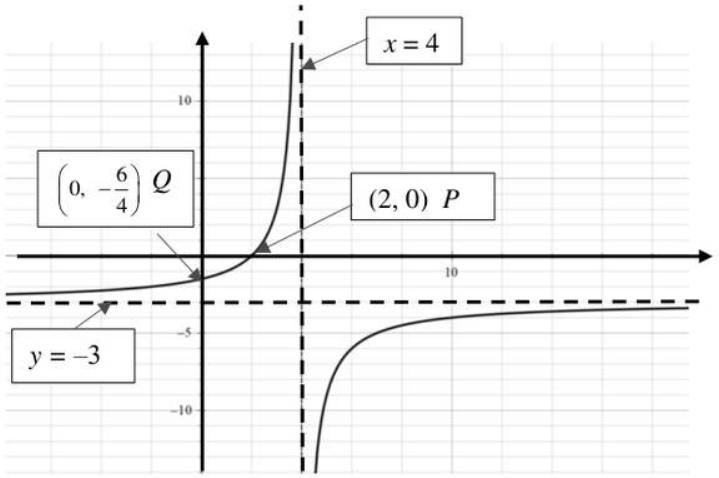
2022 JUN · Paper 02 · extract 4 of 4 · find part d in this mark scheme extract

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Type MS · Where a line meets a fraction curve again

Answer 4 of 4 · Question 10 part e · 3 marks

2023 JUN · Paper 02 · extract 1 of 3 · find part e in this mark scheme extract

Question	Scheme	Marks
10(a) **(i)**	When $y = 0$, $x = 2$ or $(2, 0)$	B1
(ii)	When $x = 0$, $y = -\frac{6}{4}$ oe or $\left(0, -\frac{6}{4}\right)$ oe	B1 [2]
	If not labelled part (i) and (ii), do not award marks unless the candidate has presented in the correct order or has made it clear which coordinate or pair of values is *P* and which is *Q*	
(b)	(i) $x = 4$ (ii) $y = -3$	B1 B1
	If not labelled part (i) and (ii), do not award marks unless the candidate has presented in the correct order or has identification of horizontal (or parallel to *x*-axis) and vertical (or parallel to *y*-axis) is clear.	[2]
(c)		B1 B1 ft B1 ft [3]
(d)	$\left(\frac{dy}{dx} =\right) \frac{-3(x-4)-(6-3x)\times 1}{(x-4)^2} = \left[\frac{6}{(x-4)^2}\right]$ oe	M1A1A1
	where $x = 2$ $\frac{dy}{dx} = \frac{6}{(2-4)^2} = \frac{6}{4} \Rightarrow$ Gradient of normal $= -\frac{4}{6}$ oe	M1
	Equation of normal: $y - 0 = "-\frac{4}{6}"(x-2) \Rightarrow [3y = -2x + 4]$ oe	dM1A1 [6]
(e)	$\frac{6-3x}{x-4} = \frac{-2x+4}{3} \Rightarrow 2x^2 - 21x + 34 = 0$	M1
	$\Rightarrow (2x-17)(x-2) = 0$	M1
	At $R \Rightarrow x = \frac{17}{2}$, (2)	A1 [3]
		Total 16 marks

Type MS · Where a line meets a fraction curve again

Answer 4 of 4 · Question 10 part e · 3 marks

2023 JUN · Paper 02 · extract 2 of 3 · find part e in this mark scheme extract

Part	Mark	Notes
(a) (i)	B1	Must state $y = 0$, $x = 2$ or $(2, 0)$ or clearly stating $x = 2$
(ii)	B1	Must state $x = 0$, $y = -\frac{6}{4}$ oe or $\left(0, -\frac{6}{4}\right)$ oe or clearly stating $y = -\frac{6}{4}$
(b)(i)	B1	For $x = 4$
(ii)	B1	For $y = -3$
(c)	B1	For a negative reciprocal curve drawn anywhere in the grid – there **must** be two branches present, they **must not** cross any asymptotes drawn and **must not** obviously 'bend back' on themselves. Mark intention.
	B1ft	For the asymptotes drawn, follow through their (b)(i) and (ii). There **must** be at least one branch of a negative reciprocal curve present in the correct place for their work in (a) and (b), which **must** not cross or obviously bend back from the asymptotes. The asymptotes **must** be labelled with their equation or shown as passing through 4 on the x-axis and -3 on the y-axis, both clearly labelled.
	B1ft	For at least one branch of a negative reciprocal curve in the correct place for their work in (a) and (b), passing through their intercepts on the axes. The intercepts should be correctly labelled with the coordinates or the axes labelled correctly with $-\frac{6}{4}$ (oe) and 2 (or their ft values) but condone labelling to be P and Q.
(d)	M1	For an expression of the form. $\frac{-a(x-4)-(6-3x)\times b}{(x-4)^2}$ oe
	A1	For an expression of the form. $\frac{-3(x-4)-(6-3x)\times b}{(x-4)^2}$ or $\frac{-a(x-4)-(6-3x)\times 1}{(x-4)^2}$ .oe
	A1	Fully correct – need not be simplified.
	M1	For substituting $x = 2$ into their $\frac{dy}{dx}$ and finding the gradient of the normal. This is not a dependent method mark, but the substitution must be into a **changed** function. If their expression does not allow substitution of $x = 2$, this mark cannot be awarded.
	dM1	For a complete and correct method to find the equation of the normal using their (changed gradient), $y = 0$ and $x = 2$. Dependent on the previous mark. If $y = mx + c$ is used they must find a value for c.
	A1	For any correct equation. **This can be in any form and may be left unsimplified.**
(e)	M1	For equating their equation of the normal to C and attempting to form a 3TQ. The attempt must involve correctly removing both denominators of the equation as a minimum and any attempt to collect terms.
	M1	For a minimally acceptable (see general guidance) and complete attempt to solve their quadratic equation, leading to a value of x..
	A1	For the x coordinate of point R $x = \frac{17}{2}$

Type MS · Where a line meets a fraction curve again

Answer 4 of 4 · Question 10 part e · 3 marks

2023 JUN · Paper 02 · extract 3 of 3 · find part e in this mark scheme extract

Question	Scheme	Marks